

LAB EXPERIMENTS:

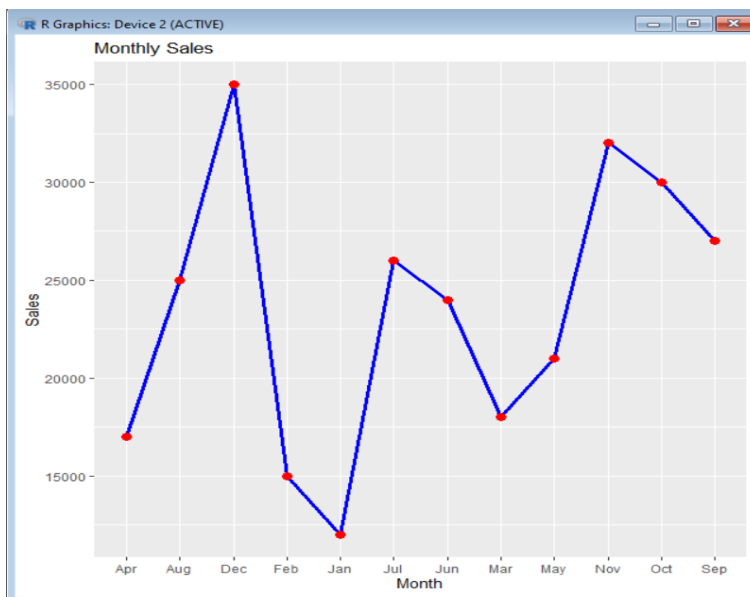
Set 1

1. Line chart to visualize the monthly sales:

Python code:

```
sales <- data.frame(  
  Month = c("Jan","Feb","Mar","Apr","May","Jun",  
            "Jul","Aug","Sep","Oct","Nov","Dec"),  
  Sales = c(12000,15000,18000,17000,21000,24000,  
            26000,25000,27000,30000,32000,35000),  
  Product = c("A","B","C","D","E","F",  
              "G","H","I","J","K","L"),  
  AdBudget = c(2000,2500,3000,2800,3500,4000,  
               4200,4500,4800,5000,5500,6000)  
)  
  
ggplot(sales, aes(x = Month, y = Sales, group = 1)) +  
  geom_line(color = "blue", size = 1.2) +  
  geom_point(size = 3, color = "red") +  
  labs(title = "Monthly Sales",  
        x = "Month",  
        y = "Sales")
```

Output:

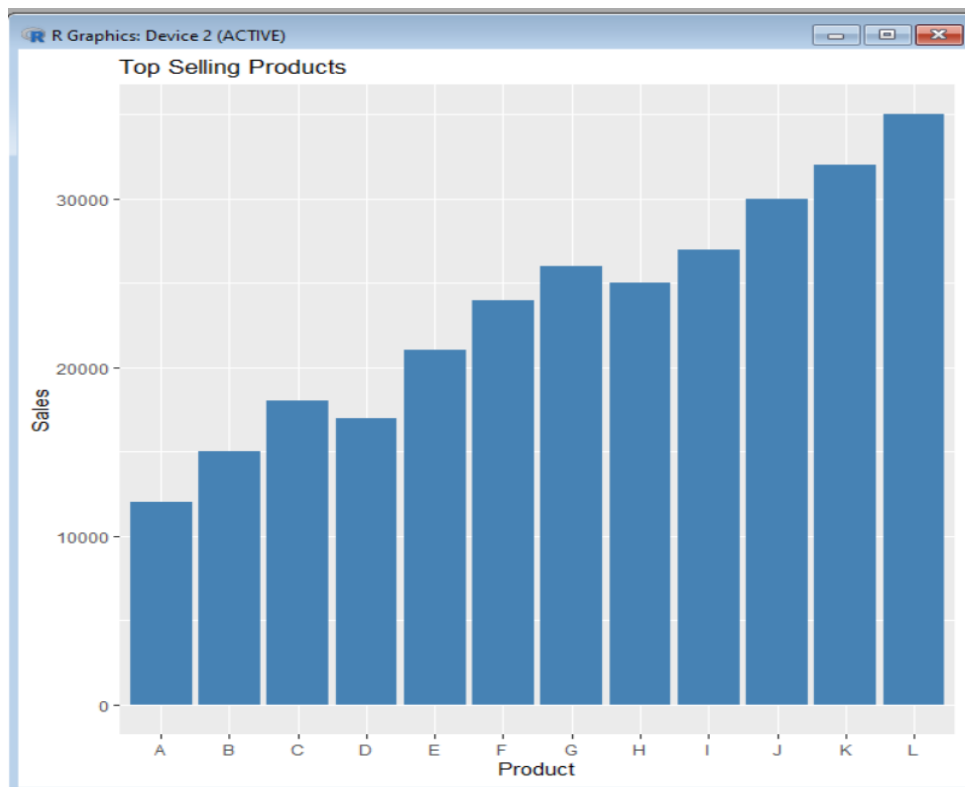


2. Top-selling products for the year:

Python code:

```
sales <- data.frame(  
  Month = c("Jan","Feb","Mar","Apr","May","Jun",  
            "Jul","Aug","Sep","Oct","Nov","Dec"),  
  Sales = c(12000,15000,18000,17000,21000,24000,  
            26000,25000,27000,30000,32000,35000),  
  Product = c("A","B","C","D","E","F",  
              "G","H","I","J","K","L"),  
  AdBudget = c(2000,2500,3000,2800,3500,4000,  
               4200,4500,4800,5000,5500,6000)  
)  
ggplot(sales, aes(x = Product, y = Sales)) +  
  geom_bar(stat = "identity", fill = "steelblue") +  
  labs(title = "Top Selling Products",  
       x = "Product",  
       y = "Sales")
```

Output:

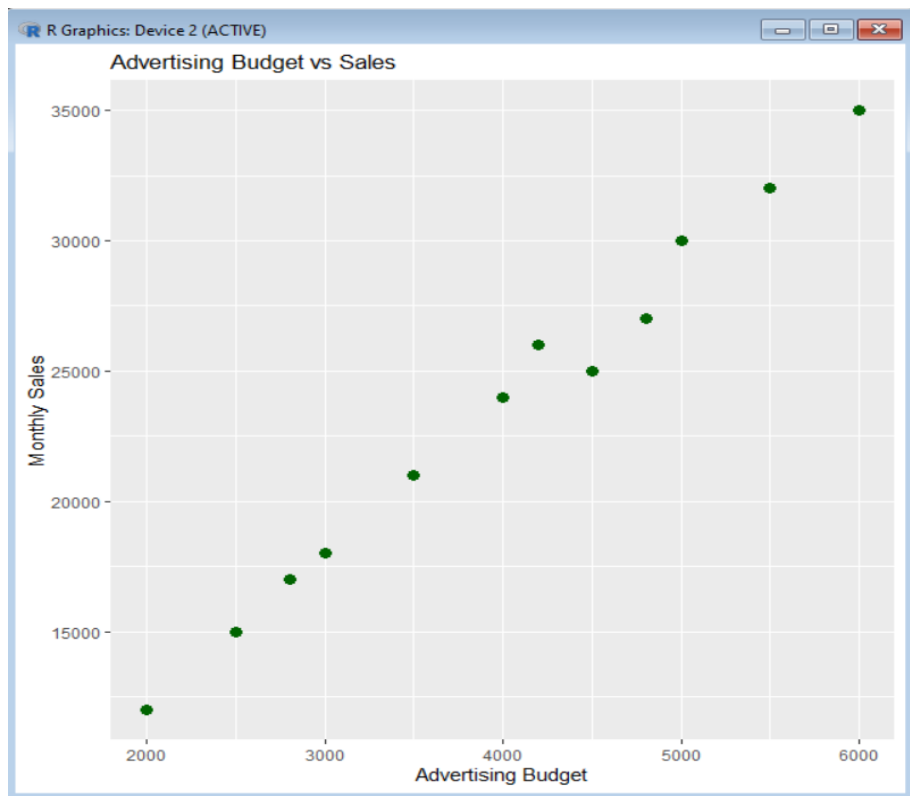


3. The relationship between advertising budget and monthly sales:

Python code:

```
sales <- data.frame(  
  Month = c("Jan", "Feb", "Mar", "Apr", "May", "Jun",  
            "Jul", "Aug", "Sep", "Oct", "Nov", "Dec"),  
  Sales = c(12000, 15000, 18000, 17000, 21000, 24000,  
            26000, 25000, 27000, 30000, 32000, 35000),  
  Product = c("A", "B", "C", "D", "E", "F",  
              "G", "H", "I", "J", "K", "L"),  
  AdBudget = c(2000, 2500, 3000, 2800, 3500, 4000,  
               4200, 4500, 4800, 5000, 5500, 6000)  
)  
  
ggplot(sales, aes(x = AdBudget, y = Sales)) +  
  geom_point(size = 3, color = "darkgreen") +  
  labs(title = "Advertising Budget vs Sales",  
       x = "Advertising Budget",  
       y = "Monthly Sales")
```

Output:



4. Combining the line chart and bar chart:

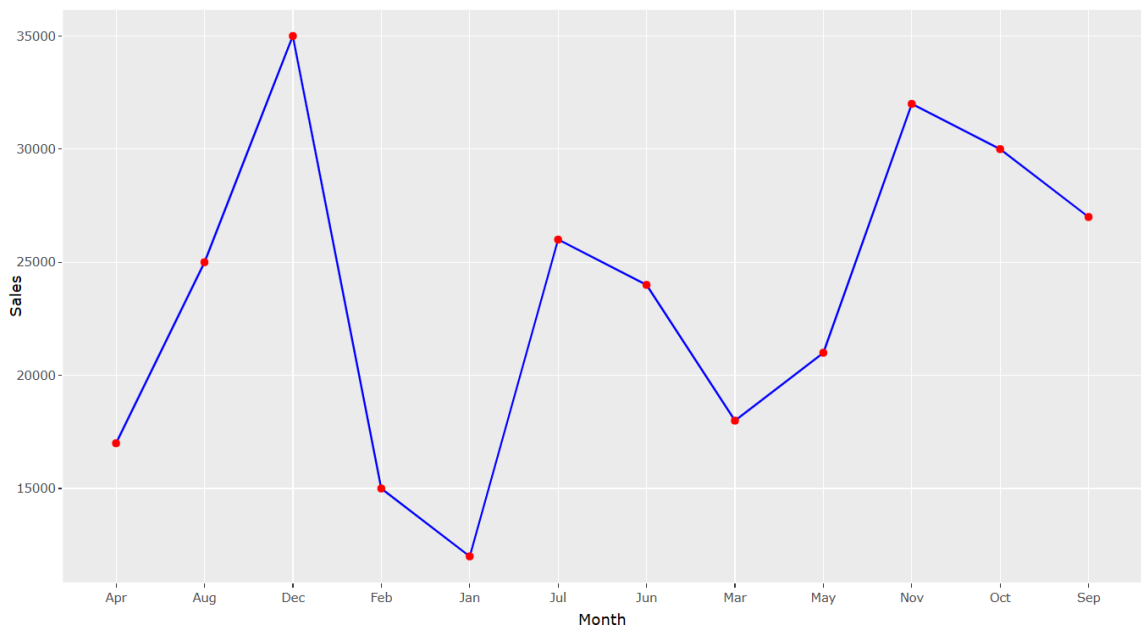
4a. Line chart:

Python code:

```
sales <- data.frame(  
  Month = c("Jan", "Feb", "Mar", "Apr", "May", "Jun",  
            "Jul", "Aug", "Sep", "Oct", "Nov", "Dec"),  
  Sales = c(12000, 15000, 18000, 17000, 21000, 24000,  
            26000, 25000, 27000, 30000, 32000, 35000)  
)  
  
p1 <- ggplot(sales, aes(x = Month, y = Sales, group = 1)) +  
  geom_line(color = "blue") +  
  geom_point(color = "red")
```

ggplotly(p1)

Output:



4b. bar chart:

Python code:

```
sales <- data.frame(  
  Product = c("A", "B", "C", "D", "E", "F",
```

```

      "G","H","I","J","K","L"),
Sales = c(12000,15000,18000,17000,21000,24000,
          26000,25000,27000,30000,32000,35000)
)

```

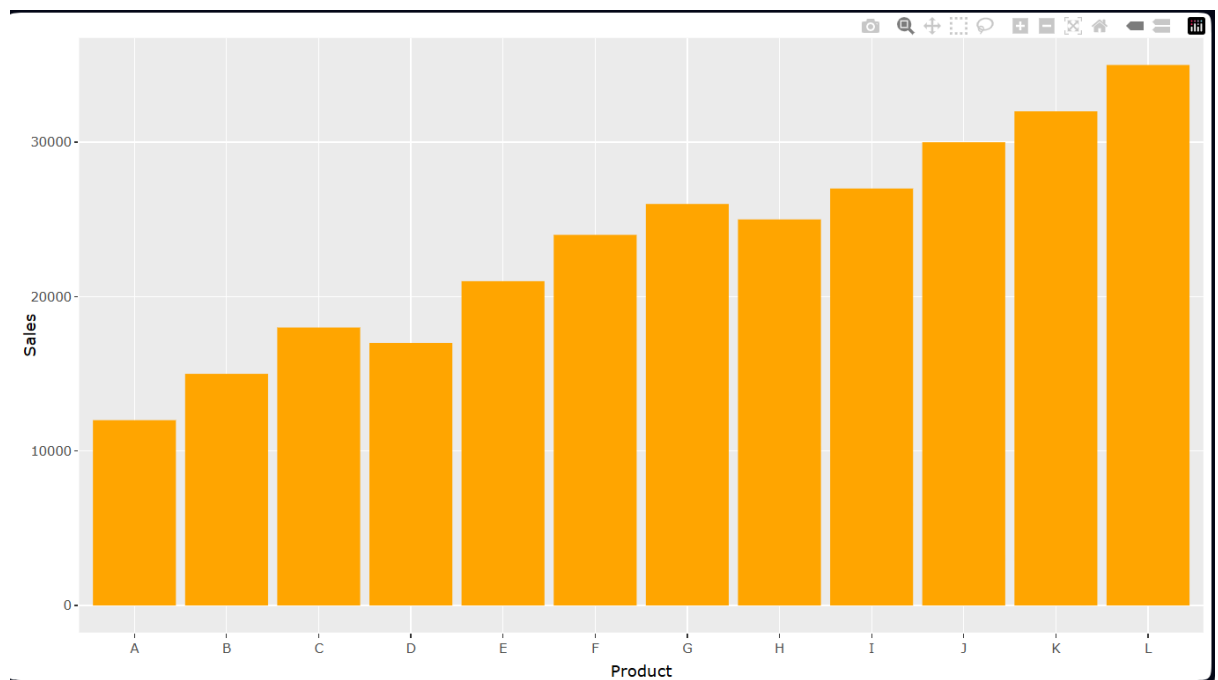
```

p2 <- ggplot(sales, aes(x = Product, y = Sales)) +
  geom_bar(stat = "identity", fill = "orange")

```

```
ggplotly(p2)
```

Output:



Set 2

1. Histogram to represent the distribution of customer ages:

Python code:

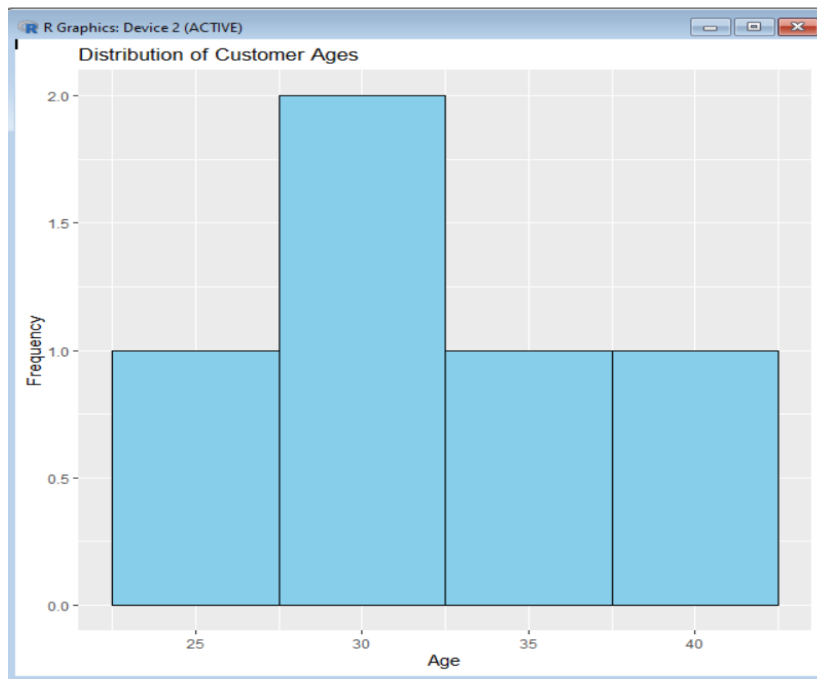
```

customer <- data.frame(
  CustomerID = c(1,2,3,4,5),
  Age = c(25,30,35,28,40),
  Satisfaction = c(4,5,3,4,5)
)

```

```
ggplot(customer, aes(x = Age)) +
  geom_histogram(binwidth = 5, fill = "skyblue", color = "black") +
  labs(title = "Distribution of Customer Ages",
        x = "Age",
        y = "Frequency")
```

Output:



2. Pie chart to display the overall distribution of customer satisfaction scores:

Python code:

```
customer <- data.frame(
  CustomerID = c(1,2,3,4,5),
  Age = c(25,30,35,28,40),
  Satisfaction = c(4,5,3,4,5)
)
pie_data <- table(customer$Satisfaction)

pie(
  pie_data,
  labels = names(pie_data),
  col = rainbow(length(pie_data)),
  main = "Customer Satisfaction Scores"
```

)

Output:



3. Bar chart to visualize the distribution of customer satisfaction scores by age group:

Python code:

```
customer$AgeGroup <- cut(
  customer$Age,
  breaks = c(20,30,40,50),
  labels = c("21-30","31-40","41-50"),
  include.lowest = TRUE
)
```

```
ggplot(customer,
  aes(x = AgeGroup,
    fill = factor(Satisfaction))) +
  geom_bar() +
  labs(title = "Customer Satisfaction by Age Group",
    x = "Age Group",
```

```
y = "Count",  
fill = "Satisfaction")
```

Output:



4. Word cloud from open-ended customer feedback to identify prevalent customer sentiments:

Python code:

```
feedback <- c(  
  "Good service",  
  "Excellent support",  
  "Friendly staff",  
  "Fast delivery",  
  "Very satisfied",  
  "Helpful service",  
  "Great experience",  
  "Quick response",  
  "Quality product",  
  "Happy customer"  
)
```



```
wordcloud(
  words = feedback,
  random.order = FALSE,
  colors = brewer.pal(8, "Dark2")
)
```

Output:



Set 3

1. Line chart to visualize the performance trend of employees over time:

Python code:

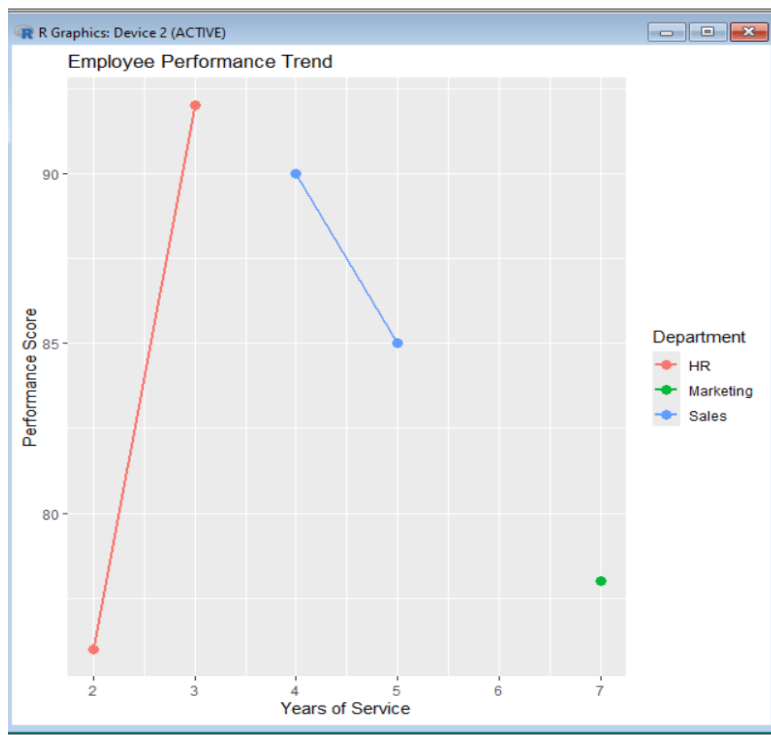
```
employee <- data.frame(
  EmployeeID = c(1,2,3,4,5),
  Department = c("Sales","HR","Marketing","Sales","HR"),
  YearsService = c(5,3,7,4,2),
  Performance = c(85,92,78,90,76)
)
ggplot(employee,
  aes(x = YearsService,
```

```

    y = Performance,
    color = Department,
    group = Department)) +
  geom_line(size = 1) +
  geom_point(size = 3) +
  labs(title = "Employee Performance Trend",
    x = "Years of Service",
    y = "Performance Score",
    color = "Department")

```

Output:



2. Bar chart showing the distribution of employees across different departments:

Python code:

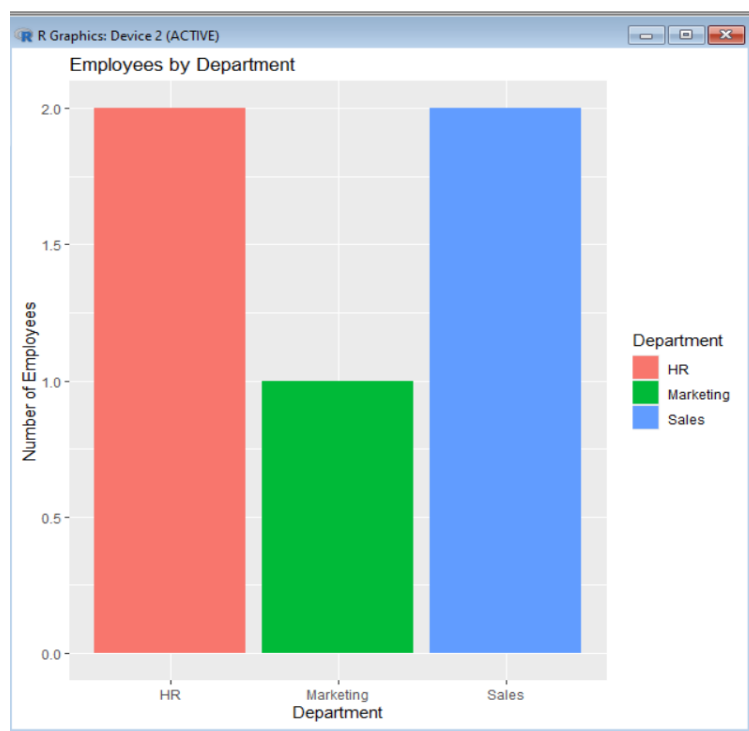
```

employee <- data.frame(
  EmployeeID = c(1,2,3,4,5),
  Department = c("Sales","HR","Marketing","Sales","HR"),
  YearsService = c(5,3,7,4,2),
  Performance = c(85,92,78,90,76)
)

```

```
)
ggplot(employee,
  aes(x = Department,
    fill = Department)) +
geom_bar() +
labs(title = "Employees by Department",
  x = "Department",
  y = "Number of Employees")
```

Output:



3. Scatter plot to analyze the correlation between years of service and performance scores:

Python code::

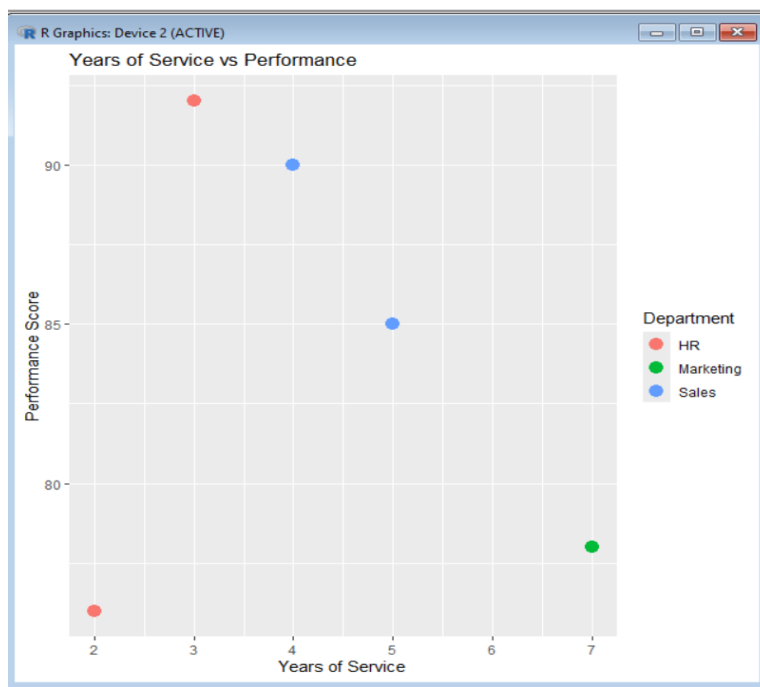
```
employee <- data.frame(
  EmployeeID = c(1,2,3,4,5),
  Department = c("Sales","HR","Marketing","Sales","HR"),
  YearsService = c(5,3,7,4,2),
  Performance = c(85,92,78,90,76)
)
ggplot(employee,
```

```

aes(x = YearsService,
    y = Performance,
    color = Department)) +
geom_point(size = 4) +
labs(title = "Years of Service vs Performance",
     x = "Years of Service",
     y = "Performance Score")

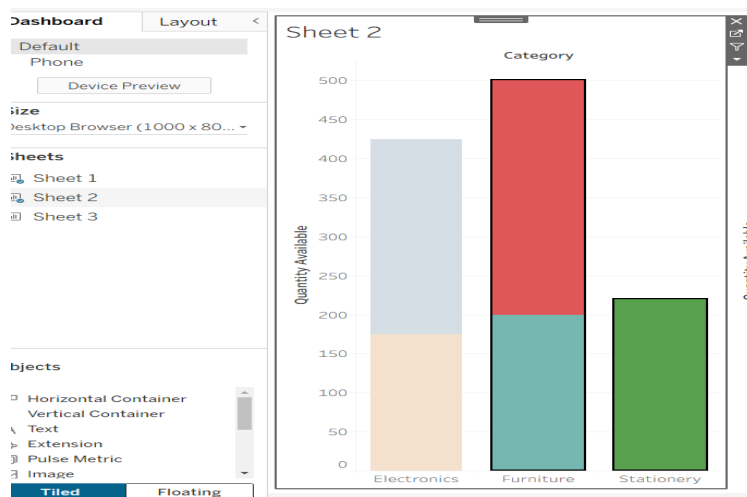
```

Output:



4. Tableau dashboard with the line chart and bar chart for interactive exploration:

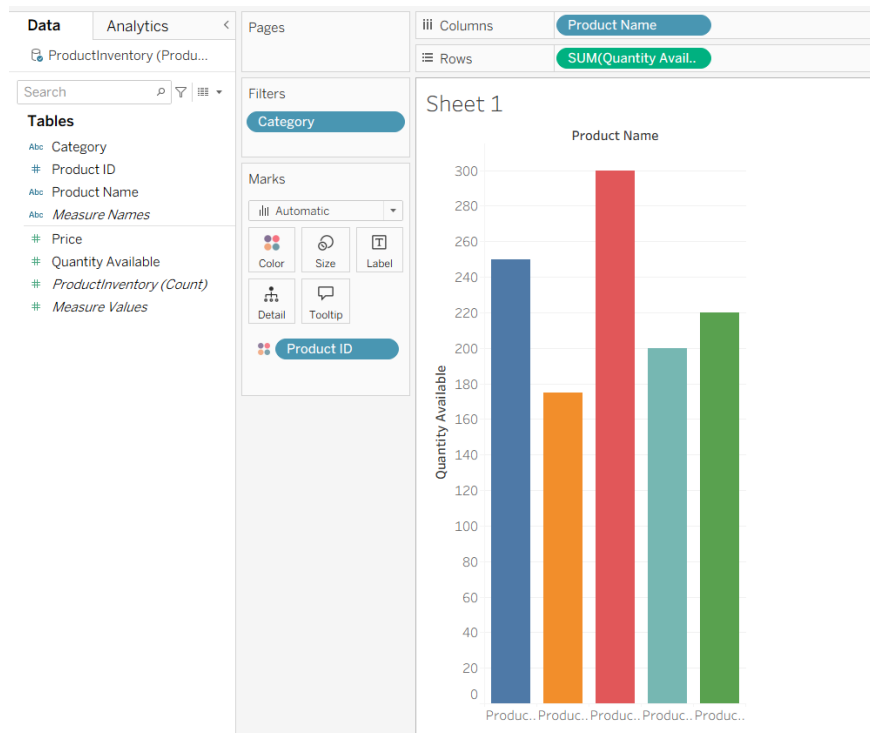
Output:



Set 4

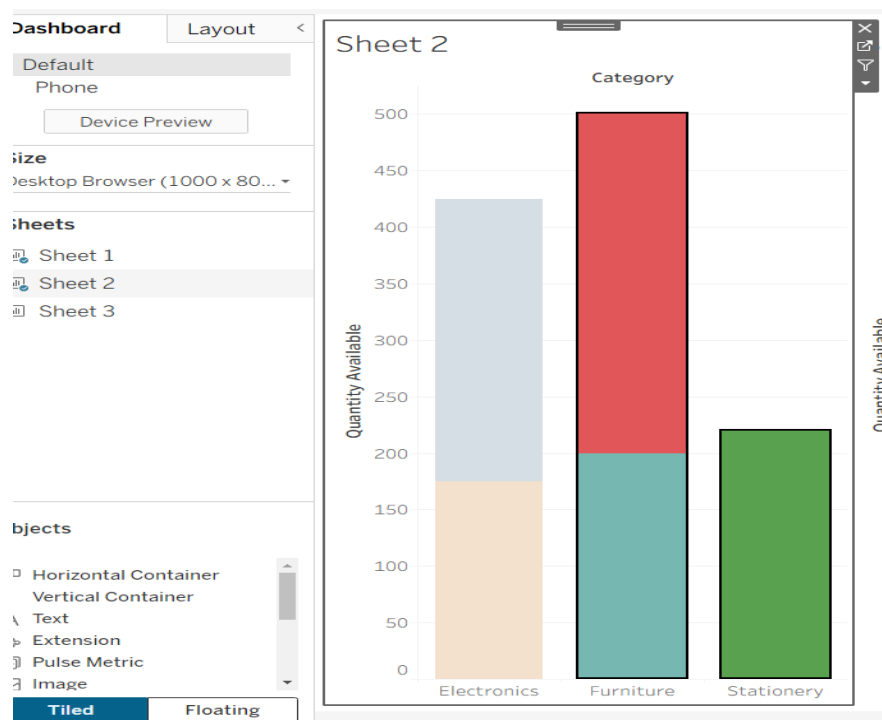
1. Bar chart to visualize the quantity of each product in the inventory:

Output:



2. stacked bar chart to show the quantity of each product within different product categories:

Output:



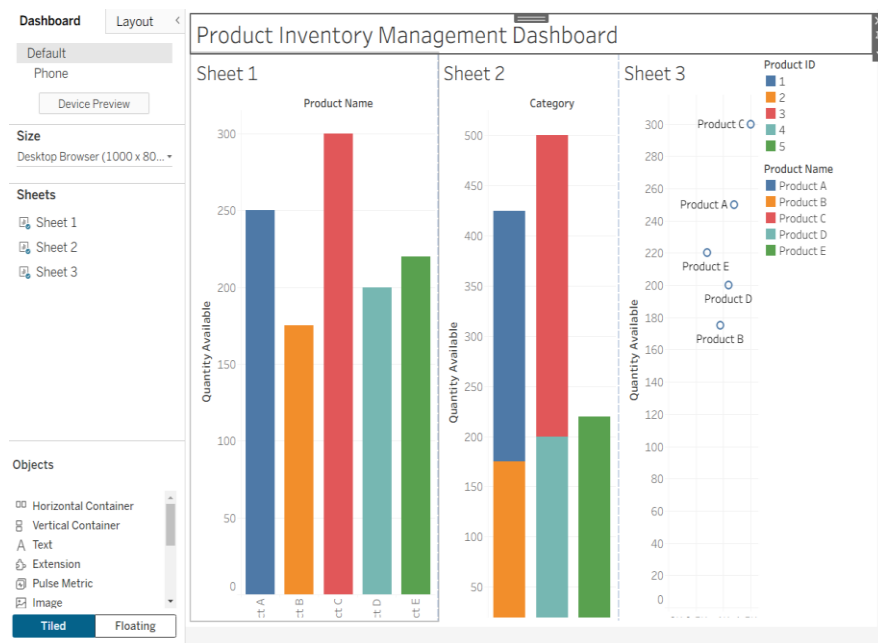
3. scatter plot to explore the relationship between product price and quantity available:

Output:



4. Tableau dashboard with the bar chart and stacked bar chart to allow users to interact with the data.

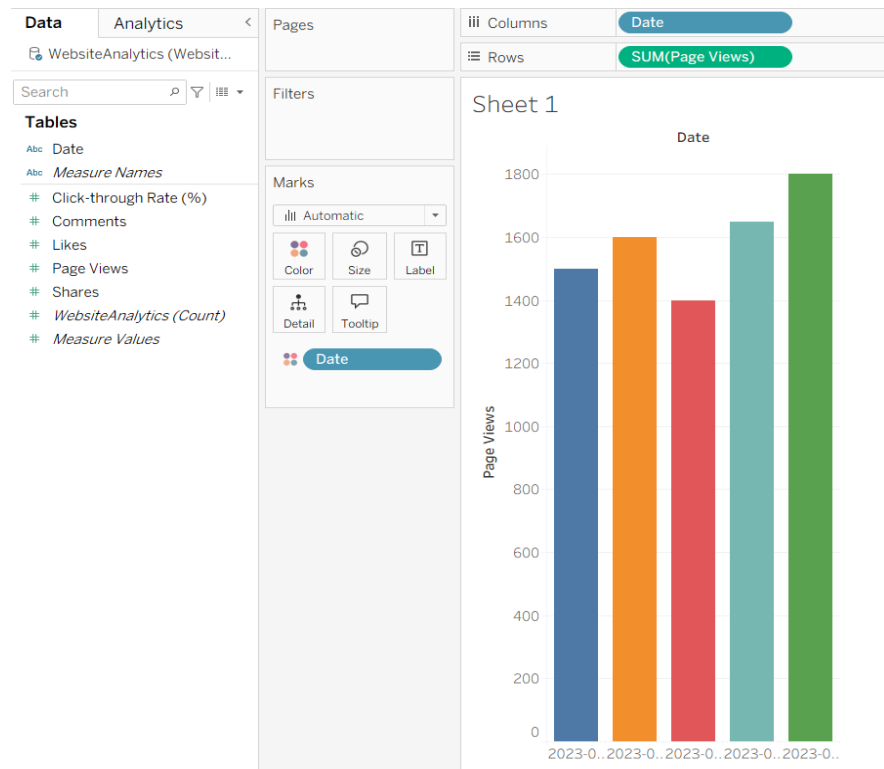
Output:



Set 5

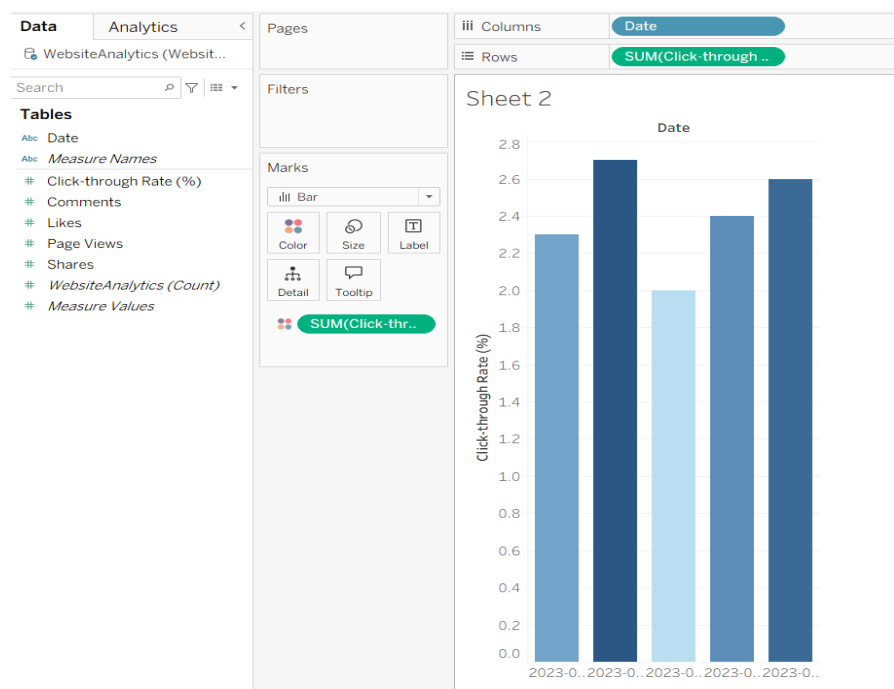
1. line chart to visualize the trend in daily page views over time:

Output:

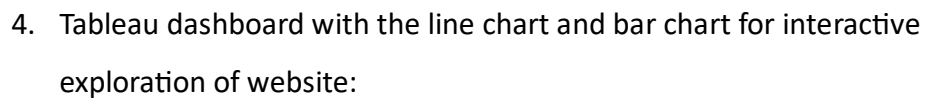


2. bar chart showing the top N days with the highest click-through rates:

Output:



- Output:



Output:

